PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T29 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 25 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T29 • Expertise Q16+ • 5 items • 25 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Sequences

Unit 2 • Expertise • 25 marks • 5 item(s)

June 2024 | 4MA1-1H Q24 | 6 marks

Nonlinear or integer-constrained series recovery

November 2023 | 4MA1-1H Q19 | 6 marks

Recover number of arithmetic terms

November 2024 | 4MA1-1H Q19 | 5 marks

Recover first term and common difference

January 2021 | 4MA1-1H Q24 | 4 marks

Symbolic relation between partial sums

May-Nov 2020 | 1H Q24 | 4 marks

Arithmetic nth term from listed terms

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q24

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°
The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.
Show clear algebraic working.

....°

WORKING SPACE

4MA1-1H Q24

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°
The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.
Show clear algebraic working.

.... °

Arithmetic sequence of polygon angles **6 marks**

Two sums

$$S_n = \frac{n}{2}[168 + 4(n-1)] = 2n^2 + 82n, \quad S_n = 180(n-2).$$

Solve

$$n^2 - 49n + 180 = 0 \Rightarrow n = 45 \ (n > 5).$$

Angle sum

$$180(45-2) = 7740.$$

Final answer

$$7740^\circ$$

4MA1-1H Q19 demand

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x
The sum of the terms of the sequence is 7260

Find the value of x
Show clear algebraic working.

$x =$ ....

WORKING SPACE

4MA1-1H Q19 demand **6 marks**

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x
The sum of the terms of the sequence is 7260

Find the value of x
Show clear algebraic working.

$x =$ ....

Last term of an arithmetic sequence **6 marks**

Demand – Last term of an arithmetic sequence

Identify a and d

$$a = 3, \quad d = 4.$$

Use the sum

$$S_n = \frac{n}{2}[2(3) + (n-1)4] = 7260 \Rightarrow 2n^2 + n - 7260 = 0.$$

Solve for the number of terms

$$(2n + 121)(n - 60) = 0 \Rightarrow n = 60.$$

Find the last term

$$x = 3 + (60 - 1)4 = 239.$$

Final answer

239

4MA1-1H Q19 M01 **5 marks**

19 An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

....

WORKING SPACE

Find the sum of an arithmetic series **5 marks**

M01 – Find the sum of an arithmetic series

Use the sum of 30 terms

$$30a + 435d = 4395 \Rightarrow 2a + 29d = 293.$$

Use the 10th and 20th terms

$$(a + 9d) + (a + 19d) = 284 \Rightarrow 2a + 28d = 284.$$

Eliminate

$$d = 9, \quad a = 16.$$

Use the sum formula

$$S_{45} = \frac{45}{2}[2(16) + 44(9)] = 9630.$$

Final answer

9630

24 An arithmetic series has first term a and common difference d.

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d.
Show your working clearly.

$a =$ ....

Worked solution

January 2021 | Question 24 | 4 marks

Family: Recover arithmetic-series parameters • Variant: Symbolic relation between partial sums

Method / working

1. Write S■n=n[2a+(2n-1)d].
2. Set S■n equal to 4S_n and cancel n.
3. Simplify to 2a-d=4a-2d.
4. Therefore a=d/2.

Final answer: a=d/2

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

....

(Total for Question 24 is 4 marks)

Answer reference | May-Nov 2020 | Question 24

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

24	$a = 8\ \ d = 7$			4	M1 can be implied
	$\left(S_{100} =\right)\dfrac{100}{2}\left(2\times 8+(100-1)\times 7\right)$ (= 35 450) **or** $\left(S_{49} =\right)\dfrac{49}{2}\left(2\times 8+(49-1)\times 7\right)$ (= 8624) **or** $\left(S_{50} =\right)\dfrac{50}{2}\left(2\times 8+(50-1)\times 7\right)$ (= 8975)				M1
	'35450' – '8624' **or** '35450' – '8975' + (8 + (50 – 1)×7)				M1
			26 826		A1
					Total 4 marks
	Alternative scheme				
	$(u_n =)\ 7n + 1$	$a = 8$ and $d = 7$		4	M1 can be implied
	$(u_{50} =)\ 7 \times 50 + 1$ (= 351) **or** $(u_{100} =)\ 7 \times 100 + 1$ (= 701)	$(u_{50} =)\ 8 + (50 – 1)\times 7$ (= 351)			M1
	$\dfrac{51}{2}('351'+'701')$	$\dfrac{51}{2}\left(2\times 351+(51-1)\times 7\right)$			M1
			26 826		A1
					Total 4 marks

Worked solution

May-Nov 2020 | Question 24 | 4 marks

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

Method

The arithmetic sequence is

$8,\ 15,\ 22,\ 29,\ 36,\ldots$

So

$a=8,\quad d=7$

The nth term is

$u_n=8+7(n-1)=7n+1$

$u_{50}=7(50)+1=351$

$u_{100}=7(100)+1=701$

There are

$100-50+1=51$

terms from the 50th to the 100th inclusive.

$sum=\frac{51}{2}(351+701)$

$=26826$

Answer: 26826